

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Statistics Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

STATISTICS TOOLKIT · CH6 · L21

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The Map

This lesson collapses Statistics Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Choose the average

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use frequency table totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Estimate the mean

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Interpret spread

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 58 website questions, 58 checked solutions, 4 local PDFs read.

01 Choose the average

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w < 3$	12
$3 < w < 4$	16
$4 < w < 5$	9
$5 < w < 6$	2
$6 < w < 7$	1

(a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) (c) One of the 40 babies is going to be chosen at random. Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

02 Use frequency table totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w < 3$	12
$3 < w < 4$	16
$4 < w < 5$	9
$5 < w < 6$	2
$6 < w < 7$	1

(a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) (c) One of the 40 babies is going to be chosen at random. Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

03 Estimate the mean

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 The table shows information about the weights, in kilograms, of 40 babies.

Weight (w kg)	Frequency
$2 < w < 3$	12
$3 < w < 4$	16
$4 < w < 5$	9
$5 < w < 6$	2
$6 < w < 7$	1

(a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) (c) One of the 40 babies is going to be chosen at random. Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

04 Interpret spread

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

5, 7, 11, 12, 13, 14, 15, 16, 17, 18, 18

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q14

Put the marks in order

5, 7, 11, 12, 13, 14, 15, 16, 17, 18, 18

Finish: 6.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

14 Max kept a record of the marks he scored in each of the 11 spelling tests he took one term. Here are his marks. 18 5 7 12 11 18 15 16 17 13 14 Find the interquartile range of the marks.

YOUR SOLUTION

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q2

Calculate statistic

$$a = 7$$

The median of four numbers

$$\frac{a + b}{2}$$

Finish: $a = 7, b = 10, c = 12$.

FORMULA BANK

Mean = $\frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

2 Given that $a < b < c$ the four whole numbers a, a, b and c have a mode of 7 a median of 8.5 a mean of 9 Work out the value of a , the value of b and the value of c . $a = b = c =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Choose the average
"work out", "show", "hence", a familiar exam phrase	Use frequency table totals
"work out", "show", "hence", a familiar exam phrase	Estimate the mean
"work out", "show", "hence", a familiar exam phrase	Interpret spread
"work out", "show", "hence", a familiar exam phrase	Calculate statistic

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.